Time Synchronisation Between Klipper Motion and Trinkey Accelerometer Data

1 Purpose

The frequency response function (FRF) identification requires the commanded input acceleration and the measured output acceleration to be expressed on the same time axis. In this experiment, the input motion is generated by Klipper, while the output acceleration is measured by two BNO055 accelerometers connected to a separate Trinkey QT2040 board. Therefore, the system contains two relevant time bases:

- Klipper *print time*, which is the time base used to schedule motion on the printer MCU.
- Trinkey time, given by the timestamp t_{us} produced by the RP2040 timer on the Trinkey.

The aim of the synchronisation procedure is to map each Trinkey timestamp to Klipper print time. After this mapping, the input and output signals can be compared sample-by-sample using a common time coordinate.

2 Klipper Print Time

Klipper plans motion on the host computer, but the final step pulses are executed by the printer MCU. Internally, Klipper uses a variable called *print time* to represent the scheduled time of a motion command. For the primary MCU, print time is directly related to the MCU clock:

$$c = f_{\mathrm{mcu}} p, \tag{1}$$

where c is the MCU clock value in ticks, f_{mcu} is the MCU clock frequency, and p is Klipper print time in seconds. Therefore, print time can be interpreted as the MCU clock expressed in seconds. Klipper's code overview documents this host/MCU clock model and the conversion between MCU clock values and print time [6].

When a vibration test is executed, each input motion segment has a known print time. If the first segment starts at print time p_0, and a generated segment starts at local test time τ_k, then its scheduled start time is

$$p_k = p_0 + \tau_k. \tag{2}$$

The corresponding input acceleration for that segment is known from the trajectory generation code. Therefore, the input acceleration can be stored as a piecewise constant signal in Klipper print time:

$$u(p) = a_k, \qquad p_k \le p < p_k + \Delta t_k, \tag{3}$$

where a_k is the acceleration of segment k, and Δt_k is its duration.

3 Trinkey Time

The Trinkey firmware records accelerometer samples with its own timestamp t_{us}, measured in microseconds. This timestamp is produced by the Trinkey RP2040 and is independent from the printer MCU clock. For analysis it is first converted to seconds:

$$q = 10^{-6} t_{\text{us}}. \tag{4}$$

The two BNO055 accelerometers are connected to the same Trinkey board. Therefore, both accelerometers share the same Trinkey time base. This is useful, because one time synchronisation model can be used for both the toolhead and bed accelerometer data.

4 Clock Model

Over the duration of one vibration sweep, the relation between Trinkey time and Klipper print time can be modelled as an affine function:

$$p = \alpha q + \beta, \tag{5}$$

where:

- q is Trinkey time in seconds,
- p is Klipper print time in seconds,
- α is the relative clock rate,
- β is the clock offset.

If both clocks ran at exactly the same rate, then $\alpha = 1$. In practice, α can be slightly different from one due to oscillator tolerances and clock drift. The offset β accounts for the fact that the two clocks do not start from the same zero time. This is the standard clock offset and clock skew model used in clock synchronisation methods, including sensor-network synchronisation approaches [3].

5 Synchronisation Measurements

To estimate α and β, the Klipper host sends repeated synchronisation requests to the Trinkey. The vibration-test logger takes 20 observations before and 20 after each excitation. Continuous streaming instead maintains a rolling window of at most 64 observations and normally requests one new observation per second. For one synchronisation request:

1. The Klipper host records its monotonic time before sending the request: $h_{0,i}$.
2. The host sends a synchronisation command to the Trinkey.
3. The Trinkey immediately replies with its current timestamp q_i.
4. The host records its monotonic time after receiving the reply: $h_{1,i}$.

The host time corresponding to the Trinkey reply is estimated by the midpoint:

$$h_i = \frac{h_{0,i} + h_{1,i}}{2}. \tag{6}$$

The round-trip time is

$$r_i = h_{1,i} - h_{0,i}. \tag{7}$$

The midpoint approximation assumes that the outward and return communication delays are approximately symmetric. This is not perfectly true for USB serial communication, but repeated measurements allow observations with additional queueing delay to be rejected. This follows the general principle of two-way clock synchronisation methods such as Cristian's algorithm and the Network Time Protocol (NTP), where round-trip delay is used to estimate and qualify a clock offset observation [1, 2].

The host monotonic time h_i is then converted to Klipper print time using Klipper's clock synchronisation estimate:

$$p_i = \text{estimated_print_time}(h_i). \tag{8}$$

Each synchronisation request therefore gives one calibration pair:

$$(q_i, p_i). \tag{9}$$

6 Least-Squares Clock Fit

The parameters α and β are estimated by fitting the model in Equation (5) to all accepted synchronisation pairs. This is an ordinary linear least-squares (OLS) problem with an intercept term [4]. The measurements can be written in matrix form as

$$p = X\theta + e, \tag{10}$$

where

$$p = \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_N \end{bmatrix}, \qquad X = \begin{bmatrix} 1 & q_1 \\ 1 & q_2 \\ \vdots & \vdots \\ 1 & q_N \end{bmatrix}, \qquad \theta = \begin{bmatrix} \beta \\ \alpha \end{bmatrix}. \tag{11}$$

Here, p is the vector of Klipper print times, q_i are the Trinkey timestamps converted to seconds, and e is the timing error in the synchronisation measurements. The first column of X is equal to one because the model includes the offset β. The least-squares estimate is obtained by minimizing the squared residual norm:

$$\hat{\theta} = \arg\min_{\theta} \|p - X\theta\|_2^2 . \tag{12}$$

The closed-form normal-equation solution is

$$\hat{\theta} = (X^T X)^{-1} X^T p, \tag{13}$$

provided that $X^T X$ is non-singular. In implementation, the inverse should not be formed explicitly. The implementation uses MATLAB's backslash operator, $\hat{\theta} = X \backslash p$, which returns a least-squares solution for an overdetermined rectangular system [5]. This avoids explicitly forming the normal-equation inverse.

After these parameters have been estimated, every accelerometer sample can be mapped to Klipper print time:

$$p_{\text{sample}} = \alpha \left(10^{-6} t_{\text{us}}\right) + \beta. \tag{14}$$

7 Latency Rejection

USB and operating system scheduling can occasionally delay a synchronisation request. Such delayed requests should not be used directly, because they reduce the accuracy of the clock fit. The round-trip time r_i is used as a quality measure.

The selected rejection rule is

$$r_i \leq r_{\min} + r_{\text{tol}}, \tag{15}$$

where $r_{\min}$ is the minimum round-trip time in the current set of observations and the tolerance is fixed to

$$r_{\text{tol}} = 1\,\text{ms}. \tag{16}$$

The OLS fit uses only the accepted index set

$$\mathcal{A} = \{i : r_i \leq r_{\min} + 1\,\text{ms}\}. \tag{17}$$

Giving preference to low-delay observations is consistent with established clock-filter practice: for example, the NTPv4 clock filter orders observations by increasing round-trip delay before selecting the synchronization sample [2, Sec. 10]. However, NTP does not prescribe the 1 ms margin used here. That value is an experiment-specific engineering threshold for this USB link and must therefore be reported as part of the method, along with the number of observations retained. It should be revalidated if the host, USB connection, firmware, or sampling workload changes.

8 Implemented Estimator

The same rejection rule and unweighted OLS model are used by the online and offline implementations:

- The live mapper is implemented in `external/klipper/klippy/extras/trinkey_accel.py`. It keeps at most 64 observations. During start-up it estimates only the offset. It enables the affine OLS fit after at least eight accepted observations span at least 5 s. A fitted clock rate outside the sanity interval $0.995 \leq \alpha \leq 1.005$ is rejected in favour of the offset-only estimate.
- Vibration-test analysis is implemented in `analysis/matlab/frf_align_signals.m`. It verifies that all 40 expected observations are present, applies the same $r_{\min} + 1\,\text{ms}$ filter, requires at least three accepted observations including at least one before and one after the excitation, and then performs the affine OLS fit.

The online start-up guard is not a different clock model. It prevents a poorly conditioned slope estimate when the accepted timestamps cover too little elapsed time. The offline data already span the complete experiment, so that guard is not needed there. The retained count, minimum RTT, RTT limit, tolerance, slope, and offset are exposed or stored as diagnostics so the fitted mapping can be audited.

9 Alignment of Input and Output Signals

After synchronisation, both the input and output are represented in Klipper print time. The input is the commanded acceleration $u(p)$, obtained from the motion segments generated by Klipper. The output is the measured acceleration $y(p)$, obtained by converting each Trinkey sample timestamp using Equation (14).

The two signals can then be compared on the same time axis:

$$u(p) \quad \text{and} \quad y(p). \tag{18}$$

For frequency-domain identification, the steady-state part of the response is selected after the initial ramp. The complex gain at angular frequency $\omega = 2\pi f$ can be estimated by fitting sine and cosine components to the measured output:

$$y(p) \approx B\sin(\omega p) + C\cos(\omega p). \tag{19}$$

The same time base must be used for the input and output. Otherwise, a timing error Δt introduces a phase error

$$\Delta\phi = -\omega\Delta t. \tag{20}$$

In degrees this is

$$\Delta\phi_{\text{deg}} = -360 f \Delta t. \tag{21}$$

This shows why time synchronisation is important. For example, a timing error of 1 ms causes a phase error of 1.8 deg at 5 Hz, but 36 deg at 100 Hz.

10 Remaining Sources of Phase Error

The proposed synchronisation method removes the main timing uncertainty caused by USB serial latency and host-side logging. However, it does not remove all possible phase errors. The following effects may still remain:

- BNO055 internal filtering and sensor group delay.
- Small timing differences between reading the bed and toolhead accelerometers, because they are read sequentially on the same I2C bus.
- Quantisation and jitter in the generated motion segments.
- Mechanical phase lag caused by the printer structure, which is the desired quantity for FRF identification.

These effects should be considered when interpreting the measured phase. In particular, any approximately constant additional delay can be estimated and reported separately from the mechanical dynamics.

11 Summary

The synchronisation procedure maps Trinkey accelerometer timestamps onto Klipper print time using repeated sync requests and a linear clock model. The main steps are:

1. Log the commanded input acceleration using Klipper print time.
2. Log each accelerometer sample with its Trinkey timestamp.
3. Collect synchronisation pairs between Trinkey time and Klipper print time.
4. Retain pairs satisfying $r_i \leq r_{\min} + 1\,\text{ms}$.
5. Fit the affine model $p = \alpha q + \beta$ by unweighted OLS using only the retained pairs.

6. Convert all accelerometer samples to Klipper print time.

7. Use the common time base for FRF estimation.

This approach follows the same principle as Klipper's built-in accelerometer handling: sample timestamps are not based on when data arrives at the host, but on a local sensor clock that is later translated to Klipper's motion time base [7].

References

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